# Abstract linear Algebra homework8

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## Problem 1

Throughout, let $A \in M_n(\mathbb{C})$ and write $A = PJP^{-1}$ in Jordan canonical form, where $J = J_1 \oplus J_2 \oplus \cdots \oplus J_r$ is a direct sum of Jordan blocks $J_i$, with $J_i$ having eigenvalue $\lambda_i$ and size $k_i$.

### (a)

**Claim.** Let $B$ be a Jordan block of size $k$ with eigenvalue $\lambda$. Then $B = \lambda \operatorname{Id} + N$, where $N$ is a $k \times k$ matrix satisfying $N^k = 0$.

**Proof.** By definition, in the standard basis $e_1, \ldots, e_k$ of $\mathbb{C}^k$, the Jordan block $B$ acts by

$$Be_1 = \lambda e_1, \qquad Be_i = e_{i-1} + \lambda e_i \quad (2 \le i \le k).$$

Define $N := B - \lambda \operatorname{Id}$. Then

$$Ne_1 = 0, \qquad Ne_i = e_{i-1} \quad (2 \le i \le k).$$

Applying $N$ repeatedly:

$$N^m e_i = e_{i-m} \quad \text{if } m < i, \qquad N^m e_i = 0 \quad \text{if } m \ge i.$$

In particular, since $i \le k$ for every basis vector $e_i$, we have $N^k e_i = 0$ for all $i$ (as $k \ge i$). Since $\{e_1, \ldots, e_k\}$ is a basis, $N^k = 0$ as an operator. ■

### (b)

**Claim.** With $B, \lambda, N$ as in part (a), and $p_B(x) = (x - \lambda)^k$ the characteristic polynomial of $B$, we have $p_B(B) = 0$.

**Proof.** By definition,

$$p_B(B) = (B - \lambda \operatorname{Id})^k = N^k,$$

using $B = \lambda \operatorname{Id} + N$ from part (a) directly to substitute $B - \lambda \operatorname{Id} = N$. By part (a), $N^k = 0$. Hence $p_B(B) = 0$. ■

## (c)

Since $J$ is block diagonal with Jordan blocks $B_1, \ldots, B_r$ (eigenvalues $\lambda_1, \ldots, \lambda_r$, sizes $k_1, \ldots, k_r$), we have

$$xI - J = \bigoplus_{i=1}^{r} (xI_{k_i} - B_i),$$

a block diagonal matrix. The determinant of a block diagonal matrix is the product of the determinants of its blocks, so

$$p_J(x) = \det(xI - J) = \prod_{i=1}^{r} \det(xI_{k_i} - B_i) = \prod_{i=1}^{r} p_{B_i}(x).$$

Now we show $p_J(J) = 0$. Since $\mathbb{C}^n = \bigoplus_i V_i$ where $V_i$ is the subspace on which $J$ acts as $B_i$, it suffices to show $p_J(J)$ vanishes on each $V_i$. On $V_i$, $J$ acts as $B_i$, so

$$p_J(J)\big|_{V_i} = \prod_{j=1}^{r} p_{B_j}(B_i).$$

Since all the factors $p_{B_j}(B_i)$ are polynomials evaluated at the same matrix $B_i$, they commute, so we may pull out the factor $j = i$:

$$p_J(J)\big|_{V_i} = \Big(\prod_{j \neq i} p_{B_j}(B_i)\Big) \cdot p_{B_i}(B_i) = \Big(\prod_{j \neq i} p_{B_j}(B_i)\Big) \cdot 0 = 0,$$

using part (b). Since this holds on every $V_i$ and these span $\mathbb{C}^n$, we conclude $p_J(J) = 0$.

## (d)

Since $A = PJP^{-1}$,

$$p_A(x) = \det(xI - A) = \det(xI - PJP^{-1}) = \det\left(P(xI - J)P^{-1}\right) = \det(xI - J) = p_J(x).$$

So $p_A = p_J$ as polynomials. Writing $p_A(x) = \sum_i a_i x^i$, and using $A^i = PJ^iP^{-1}$,

$$p_A(A) = \sum_i a_i A^i = \sum_i a_i P J^i P^{-1} = P\Big(\sum_i a_i J^i\Big)P^{-1} =$$

$$P\, p_A(J)\, P^{-1} = P\, p_J(J)\, P^{-1} = P \cdot 0 \cdot P^{-1} = 0,$$

using part (c). Hence $A$ satisfies its own characteristic polynomial. ■

# Problem 2

## (b) first: the bijection with partitions of $n$

Let $A$ be a complex $n \times n$ matrix all of whose eigenvalues equal 1. Since $\mathbb{C}$ is algebraically closed, $A$ has a Jordan canonical form $J$, and since 1 is the only eigenvalue, $J$ is a direct sum of Jordan blocks $J_{k_1}(1), \ldots, J_{k_m}(1)$ each with eigenvalue 1, where the sizes satisfy $k_1 + \cdots + k_m = n$ (the sizes account for all $n$ dimensions). Order them $k_1 \geq k_2 \geq \cdots \geq k_m \geq 1$; then $(k_1, \ldots, k_m)$ is a partition of $n$.

**Note on terminology:** since neither side here is a group, there isn't a "homomorphism" to speak of — what we're constructing is a **bijection of sets**

$$\Phi : \{\text{conjugacy classes of } A \in M_n(\mathbb{C}) \text{ with all eigenvalues } 1\} \longrightarrow \{\text{partitions of } n\},$$

$$\Phi([A]) = (\text{sizes of the Jordan blocks of } A, \text{ arranged in decreasing order}).$$

We check $\Phi$ is well-defined, injective, and surjective.

**Well-defined.** The Jordan canonical form of $A$ is unique up to reordering of the blocks along the diagonal. So although $J$ itself is only unique up to permuting blocks, the *multiset* of block sizes — equivalently, the sorted tuple $(k_1 \geq \cdots \geq k_m)$ — is a genuine invariant of the conjugacy class $[A]$. Hence $\Phi([A])$ does not depend on which representative $A$ or which ordering of $J$ we chose.

**Injective.** Suppose $\Phi([A]) = \Phi([A']) = (k_1, \ldots, k_m)$. Then the Jordan forms of $A$ and $A'$, both arranged with blocks in the same order $k_1 \geq \cdots \geq k_m$, are the *same* matrix $J$. Since $A = P_1 J P_1^{-1}$ and $A' = P_2 J P_2^{-1}$ for some invertible $P_1, P_2$, we get

$$A' = P_2 P_1^{-1} A P_1 P_2^{-1},$$

so $A$ and $A'$ are conjugate, i.e. $[A] = [A']$.

**Surjective.** Given any partition $(k_1 \geq \cdots \geq k_m)$ of $n$, let

$$J = J_{k_1}(1) \oplus J_{k_2}(1) \oplus \cdots \oplus J_{k_m}(1),$$

the block-diagonal matrix built from Jordan blocks of these sizes, each with eigenvalue 1. Then $J$ is an $n \times n$ matrix with all eigenvalues equal to 1, and by construction $\Phi([J]) = (k_1, \ldots, k_m)$.

Since $\Phi$ is well-defined, injective, and surjective, it is a bijection. ■

## (a) Count for $n = 7$

By part (b), the conjugacy classes of $7 \times 7$ complex matrices with all eigenvalues equal to 1 are in bijection with partitions of 7. Listing them:

$$7,\ 6{+}1,\ 5{+}2,\ 5{+}1{+}1,\ 4{+}3,\ 4{+}2{+}1,\ 4{+}1{+}1{+}1,\ 3{+}3{+}1,$$

$$3{+}2{+}2,\ 3{+}2{+}1{+}1,\ 3{+}1{+}1{+}1{+}1,\ 2{+}2{+}2{+}1,\ 2{+}2{+}1{+}1{+}1,$$

$$2{+}1{+}1{+}1{+}1{+}1,\ 1{+}1{+}1{+}1{+}1{+}1{+}1.$$

That is $p(7) = \boxed{15}$ conjugacy classes.

# Problem 12

We want an operator $T$ on $\mathbb{C}^4$ with

$$p_T(z) = (z-1)(z-5)^3, \qquad m_T(z) = (z-1)(z-5)^2.$$

**Reading off the Jordan structure:**

- Eigenvalue 1: algebraic multiplicity 1 (exponent in $p_T$), and appears to the first power in $m_T$ — so there is a single Jordan block of size 1.

- Eigenvalue 5: algebraic multiplicity 3 (exponent in $p_T$), but the largest Jordan block has size 2 (exponent in $m_T$). Since the block sizes for eigenvalue 5 must sum to 3 with largest block 2, the only option is blocks of size 2 and 1.

So the Jordan form is

$$J = J_1(1) \oplus J_2(5) \oplus J_1(5) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 1 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

Take

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Computing $A = PJP^{-1}$ gives

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -4 & 5 & 1 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

Equivalently, define $T : \mathbb{C}^4 \to \mathbb{C}^4$ by

$$T(x_1, x_2, x_3, x_4) = \big(x_1,\ -4x_1 + 5x_2 + x_3,\ 5x_3,\ 5x_4\big).$$

Since $A$ is similar to $J$ (via $P$), it automatically has the same characteristic and minimal polynomials:

$$p_A(z) = p_J(z) = (z-1)(z-5)^3, \qquad m_A(z) = m_J(z) = (z-1)(z-5)^2.$$

So $T$, represented by $A$ in the standard basis, is a genuine (non-block-diagonal) example of the desired operator.

# Check-In

I want to join my friend's birthday party and go boating!